

# Breaking Down the Buzz Around Quantum Computing

Quantum computing helps scientists and researchers in solving problems above a certain complexity. Quantum computers derive their power by utilising quantum mechanics and marvels such as superposition and entanglement, which allows them to perform a variety of computational tasks exponentially faster than classical computers.

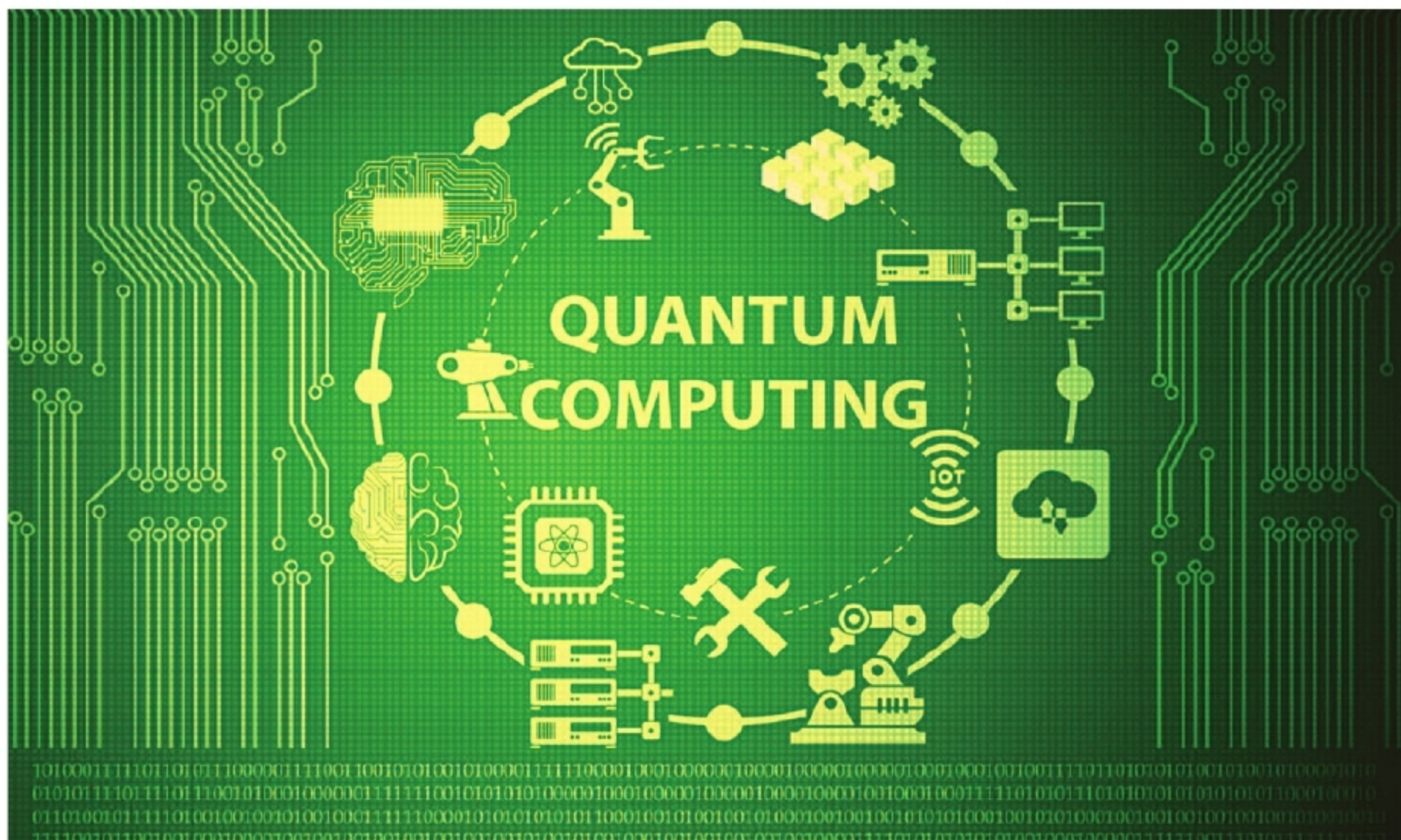
The invention of the computer has undeniably been one of the biggest technological revolutions in the history of mankind. However, classical computing is not the only way that was formulated in the last century to solve complex problems. While it was in 1927 that physicist Heisenberg introduced the uncertainty principle, it was not until 1970 that using quantum mechanics as a communication resource and the term quantum information theory came into being. So, the question

arises: how is quantum computing different from classical computing, and how did it go from being a purely theoretical subject to being used by companies like Google and IBM?

## Quantum computers versus classical computers

We already have computers and even supercomputers for faster processing speeds, then why do we need quantum computers? To understand this we need to understand the difference between the two.

A classical computer's main purpose is to save and manipulate data for working. Its chip uses bits to store this information. These bits are like tiny switches with two states—on and off, represented by one and zero, respectively. From every pixel in an image to the texts exchanged between people, everything is ultimately made up of these bits, a language that the computer understands. But even supercomputers cannot define the uncertain state that exists between on and off, especially on atomic and



(Credit: [www.eweek.com](http://www.eweek.com))

